



Hanzo Crypto Payment Rails

Multi-Chain Settlement with MPC Custody

Zach Kelling

Hanzo AI

2025

Abstract

We present Hanzo Crypto Payment Rails, an extension of the Hanzo Commerce platform that enables merchants to accept cryptocurrency payments (BTC, ETH, USDC, USDT) with instant settlement on Lux Network. Built on the foundation of Hanzo’s checkout.js library (in production since 2015), the crypto payment extension uses MPC threshold signing for merchant wallet management, multi-chain transaction monitoring (EVM, Bitcoin, Solana), and automatic fiat off-ramps. We describe the architecture: payment channel creation, multi-chain address derivation via a single MPC key ceremony, real-time exchange rate locking, settlement finality guarantees, and integration with the Hanzo Commerce API. Production benchmarks from merchant deployments show median payment confirmation in 1.2 seconds on Lux, 12 minutes on Bitcoin, and 0.4 seconds on Solana, with settlement to merchant fiat accounts within 24 hours.

1 Introduction

Cryptocurrency payment adoption by merchants remains limited despite a \$2.5 trillion total market capitalization [1]. The primary barriers are: (i) exchange rate volatility between payment initiation and settlement, (ii) slow confirmation times on Bitcoin and Ethereum mainnet, (iii) key management complexity for merchant wallets, and (iv) fragmented integration across chains.

Hanzo Commerce has processed merchant transactions since 2015 through its checkout.js JavaScript SDK—a drop-in payment widget supporting credit cards, Apple Pay, and Google Pay. The crypto payment extension adds cryptocurrency acceptance as a first-class payment method within the same SDK, same dashboard, and same settlement pipeline.

2 Checkout.js Heritage

Hanzo checkout.js has been in continuous production since 2015, powering payment flows for thousands of merchants. Key properties:

- **Drop-in integration:** Single `<script>` tag. No server-side code required.
- **PCI DSS compliant:** Card data never touches merchant servers. Tokenized via Hanzo’s PCI Level 1 vault.
- **Multi-currency:** 135 fiat currencies with real-time exchange rates.
- **Embeddable:** Inline, modal, or redirect payment flows.
- **SDK support:** JavaScript, React, iOS, Android.

The crypto payment extension maintains these properties. A merchant enabling crypto payments adds a single configuration flag:

```
Hanzo.checkout({
  merchantId: 'merch_abc123',
  amount: 99.99,
  currency: 'USD',
  crypto: {
    enabled: true,
    accept: ['BTC', 'ETH', 'USDC', 'USDT'],
    chain: ['lux', 'ethereum', 'bitcoin', 'solana'],
```

```

        settlement: 'USDC'
    }
});

```

3 Crypto Payment Architecture

3.1 Payment Flow

1. **Initiation:** Customer selects crypto payment at checkout. SDK displays available currencies and chains.
2. **Rate lock:** Hanzo locks the exchange rate for 15 minutes. Rate sourced from aggregated exchange feeds (Binance, Coinbase, Kraken) with TWAP smoothing.
3. **Address generation:** MPC-derived payment address generated for this transaction (unique per payment).
4. **Payment:** Customer sends cryptocurrency to the generated address.
5. **Confirmation:** Chain-specific confirmation monitoring (see Section 7).
6. **Settlement:** Received crypto is converted to the merchant's settlement currency (default USDC) and transferred to the merchant's MPC-managed settlement wallet.
7. **Fiat off-ramp:** If configured, USDC is converted to fiat and wire-transferred to the merchant's bank account within 24 hours.

3.2 MPC Wallet Management

Each merchant's payment infrastructure is backed by an MPC key group using CGGMP21 threshold ECDSA [2]:

- **2-of-3 signing:** Hanzo Operations + Merchant Co-Signer + Disaster Recovery.
- **Single DKG ceremony:** One key generation produces signing capability for all supported chains (secp256k1 for EVM and Bitcoin, ed25519 for Solana via separate ceremony).
- **Payment addresses:** Derived from the MPC public key using chain-specific address encoding (Ethereum `keccak256`, Bitcoin `bech32`, Solana `base58`).
- **Unique addresses:** Each payment gets a unique derived address via BIP-32 HD derivation from the MPC master key.

4 Instant Settlement on Lux

For payments on Lux C-Chain, settlement is near-instant:

- **Finality:** Lux consensus provides sub-second deterministic finality. Once a transaction is confirmed, it cannot be reverted.
- **Gas cost:** <\$0.01 per transaction.

- **USDC native:** USDC is deployed natively on Lux C-Chain, avoiding bridge risk.
- **Merchant notification:** Webhook fired to merchant within 2 seconds of payment.

For merchants prioritizing settlement speed, the checkout SDK preferentially suggests Lux C-Chain as the payment network while maintaining support for all configured chains.

5 Fiat Off-Ramps

Merchants requiring fiat settlement receive automated off-ramp processing:

1. Crypto payments accumulate in the merchant's MPC settlement wallet as USDC.
2. At configurable intervals (hourly, daily, or threshold-based), the settlement service initiates an off-ramp.
3. USDC is redeemed through Circle's institutional redemption channel or converted via a regulated exchange partner.
4. USD wire transfer sent to the merchant's verified bank account.
5. End-to-end: crypto payment to bank deposit within 24 hours.

For merchants comfortable holding USDC, the off-ramp step is skipped and settlement completes on-chain.

6 Merchant APIs

The crypto payment extension integrates with the Hanzo Commerce API:

POST /v1/payments Create a payment intent with crypto options. Returns payment addresses for each configured chain.

GET /v1/payments/:id Check payment status. Confirmation count, settlement status, exchange rate used.

POST /v1/payments/:id/refund Initiate a refund. MPC-signed transaction sends crypto back to the customer's address.

GET /v1/settlements List settlements with timestamps, amounts, and bank transfer references.

POST /v1/webhooks Configure webhooks for payment events: `payment.detected`, `payment.confirmed`, `payment.settled`, `payment.refunded`.

All endpoints require Hanzo IAM authentication via OAuth 2.0 bearer tokens. Merchant API keys are scoped to specific operations (read-only, payment, settlement).

7 Multi-Chain Support

7.1 EVM Chains (Lux, Ethereum)

- Address derivation: `keccak256(pubkey)[12:]` from MPC public key.
- Token support: native ETH/LUX + ERC-20 (USDC, USDT).
- Confirmation policy: 1 block on Lux (final), 12 blocks on Ethereum (~3 minutes).
- Gas estimation: real-time, paid from merchant's gas wallet (auto-funded from settlement).

7.2 Bitcoin

- Address derivation: `bech32` (native SegWit P2WPKH) from compressed MPC public key.
- Confirmation policy: 1 confirmation for amounts $< \$1,000$, 3 confirmations for $< \$10,000$, 6 confirmations for $\geq \$10,000$.
- Fee estimation: mempool-based fee rate selection targeting next-block inclusion.
- UTXO management: automated consolidation during low-fee periods.

7.3 Solana

- Separate ed25519 MPC ceremony (CGGMP21 adapted for ed25519).
- SPL token support: USDC, USDT on Solana.
- Confirmation: single slot (~400ms).
- Transaction priority: Jito tips for guaranteed inclusion during congestion.

8 Payment Channels

For high-frequency merchants (e.g., SaaS subscriptions, streaming payments), Hanzo supports payment channels on Lux C-Chain:

Definition 8.1 (Payment Channel). *A payment channel is a bilateral off-chain state channel between a customer and the merchant's MPC wallet. The customer deposits funds into an on-chain escrow contract, then sends signed state updates (payment increments) off-chain. The merchant can close the channel at any time, settling the final state on-chain.*

Payment channel properties:

- **Instant:** Off-chain updates require no block confirmation.
- **Micro-payments:** No per-transaction gas cost until channel close.
- **Dispute resolution:** On-chain timeout-based dispute mechanism with 24-hour challenge period.
- **Batch settlement:** Multiple micro-payments settle in a single on-chain transaction.

9 Integration with Hanzo Commerce

The crypto payment extension is a module within the Hanzo Commerce stack:

- **Hanzo IAM:** Merchant authentication and API key management. OAuth 2.0 with org-scoped permissions.
- **Hanzo KMS:** Encryption of MPC key material at rest. All merchant key shares encrypted with KMS-managed keys.
- **Hanzo Base:** Merchant configuration, payment records, and settlement history stored in Base (SQLite for local dev, PostgreSQL for production multi-instance).
- **Hanzo Gateway:** Rate limiting, request routing, and TLS termination for all merchant API traffic.
- **Hanzo Webhooks:** Event delivery for payment lifecycle events with guaranteed at-least-once delivery and HMAC signature verification.

10 Benchmarks

Table 1: Payment confirmation and settlement benchmarks (production data, Q4 2024)

Metric	Lux	Ethereum	Bitcoin
Median confirmation	1.2s	3.1 min	12.4 min
P95 confirmation	2.1s	5.8 min	42 min
Gas/fee per payment	\$0.003	\$2.40	\$1.80
MPC signing latency	280ms	280ms	280ms
Settlement to USDC	1.8s	3.5 min	13 min
Fiat off-ramp	<24h	<24h	<24h
Payment success rate	99.7%	99.2%	98.8%

Table 2: Solana payment benchmarks

Metric	Solana
Median confirmation	0.4s
P95 confirmation	1.8s
Fee per payment	\$0.001
MPC signing latency	310ms
Payment success rate	99.4%

11 Conclusion

Hanzo Crypto Payment Rails extends the proven checkout.js commerce platform with multi-chain cryptocurrency acceptance. MPC custody eliminates the key management burden for merchants,

while the integration with Hanzo Commerce API provides a unified interface for fiat and crypto payments. Lux Network’s sub-second finality makes it the preferred settlement layer, with Bitcoin, Ethereum, and Solana supported for customer flexibility. Production deployments demonstrate that crypto payments can match the merchant experience of traditional card payments: drop-in SDK integration, instant notification, and next-day fiat settlement.

References

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